



AAE4203 Laboratory Report

GNSS Data Collection and Processing

Course: AAE4203 — Guidance and Navigation
Group: Group 10
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Abstract

This laboratory report presents the implementation and analysis of a Weighted Least Squares Single Point Positioning (WLSE-SPP) algorithm for GNSS positioning in a light urban canyon environment. Data collection was conducted using a Ublox ZED-F9P receiver in the vicinity of Block X in PolyU, followed by comprehensive analysis of satellite visibility, signal quality, and positioning performance. The results demonstrate significant positioning degradation due to multipath and NLOS effects, with mean 2D and 3D errors of 30.678 m and 52.863 m respectively. The report then identifies key error sources and proposes mitigation strategies including advanced weighting schemes and sensor fusion for improved urban navigation.

Keywords: GNSS, SPP, weighted least squares, positioning degradation

1 Data Collection

1.1 Experiment Location

The experiment is conducted in the campus of The Hong Kong Polytechnic University, in vicinity of Block X. It is a light urban canyon environment, which allows sufficient open sky satellite reception while multipath, non-line-of-sight (NLOS), or diffraction degradations may be introduced by the surrounding buildings.



Figure 1: Data collection environment near Block X.

1.2 Experiment Setup

In this experiment, raw GNSS observations were collected using a commercial-grade Ublox ZED-F9P receiver paired with an ANN-MB-00 L1/L5 multi-band antenna. After connecting the antenna to the receiver and powering on the receiver through its "Power+GPS" port, the data stream was logged onto the computer through the U-Center software.



(a) Device setup



(b) Ublox F9P receiver

Figure 2: Experiment device setup.

1.3 Experiment Trajectory

The raw GNSS observation data were collected along a dynamic trajectory circling the data collection site in slow walking speed. The data incorporates two loops along the set trajectory.

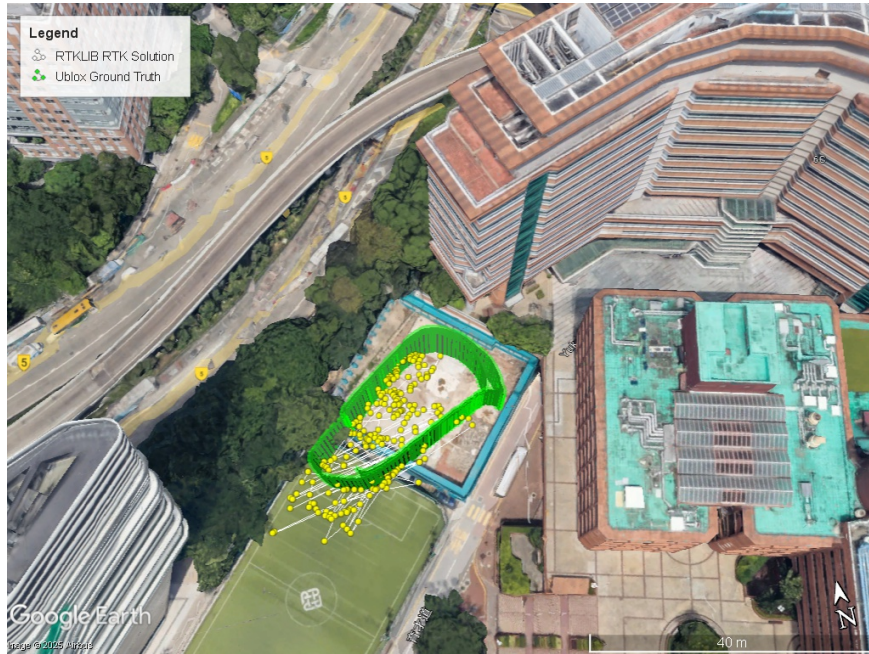


Figure 3: Data collection trajectory: ground truth output vs RTK SPP solution.

Table 1 summarizes the details of this experiment.

Table 1: Data collection summary.

Field	Value
Location	Vicinity of Block X
Conditions	Light urban canyon; partial obstruction
Duration	≈ 4 minutes @ 1 Hz
Receiver	Ublox ZED-F9P
Antenna	ANN-MB-00 L1/L5 multi-band

2 Raw Data Analysis

2.1 Number of Satellite Observations

The collected raw data are processed through RTKLIB and the rinex2csv.m script with an dependency on matRTKLIB [1]. Figure 4 shows the skyview plots of satellite received at (a) a relatively open sky location and (b) a partially obstructed location, such that scenario (a) has relatively more satellite received compared to that of scenario (b).

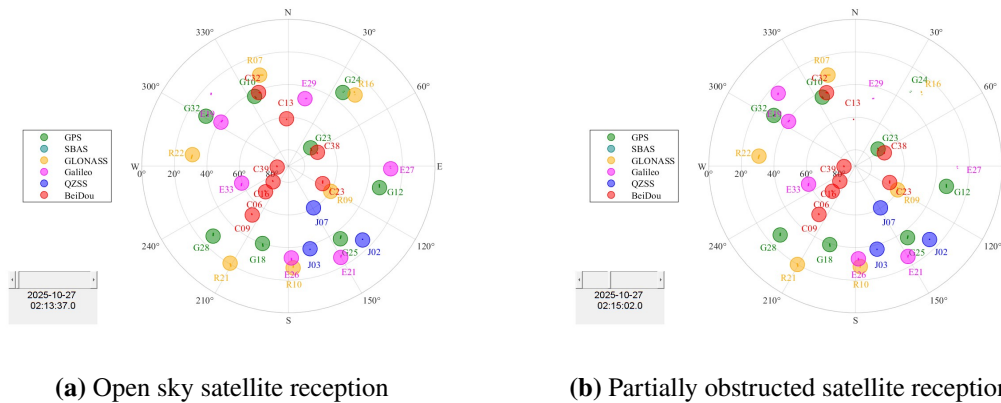


Figure 4: Skyview plots of received satellites in (a) an open sky and (b) a partially obstructed scenario.

Therefore, along the dynamic trajectory, various blockage scenarios could be observed, such that at some time instants the received L1 observations are significantly less than that of other time instants. Such phenomena is illustrated in Figure 5, where a significant decrease in L1 signal reception could be observed near the 02:15:00 and 02:17:00 timestamp.

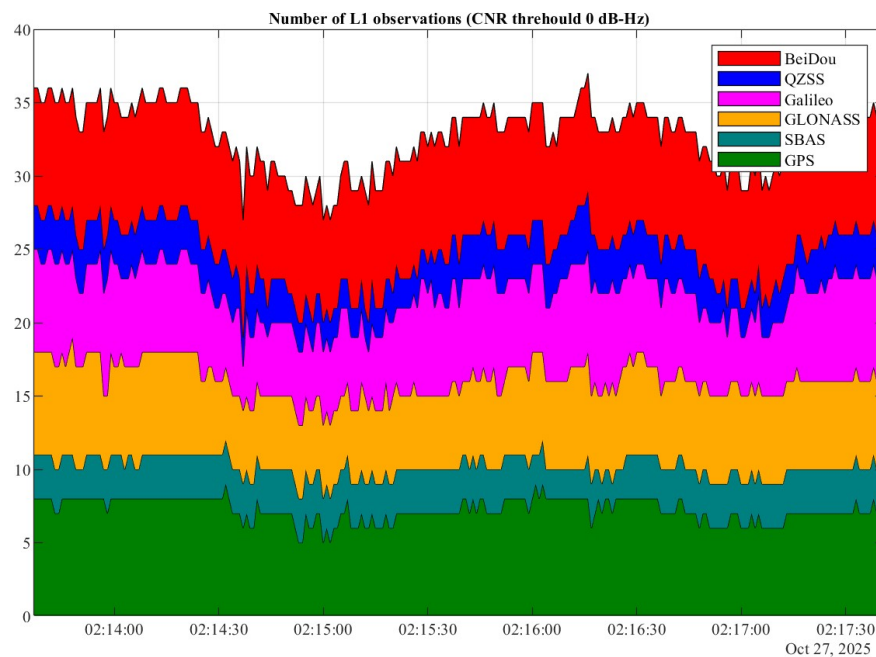


Figure 5: Number of L1 observations in the time domain.

2.2 Signal Power

Upon further investigation, it could be observed that multiple cycle slippage occurred near the identified timestamps of decreased satellite visibility, where cycle slip refers to a loss of signal that the receiver was originally tracking [2]. Additionally, as shown in Figure 6, the signal powers (SNRs) nearby the cycle slip along the temporal axis tend to be significantly attenuated. This indicates that the signal slippage occurred due to building obstructions, where the significant

attenuation before or after the signal slippage is caused by the diffraction of satellite line-of-sight (LOS) signal near building edges [3].

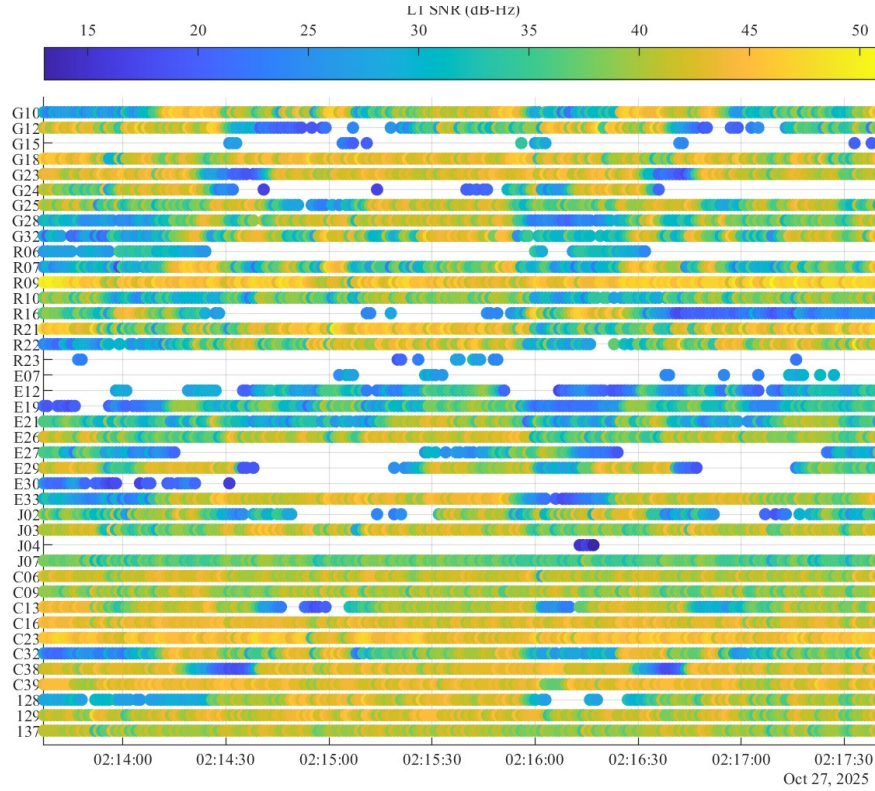


Figure 6: L1 signal to noise ratio throughout the observation period.

3 SPP Algorithm (Weighted Least Squares)

The Single Point Positioning (SPP) algorithm estimates receiver coordinates and clock bias using pseudorange measurements from multiple GNSS satellites. The fundamental pseudorange observation model for satellite i is given by [2]:

$$\rho_i = \|\mathbf{x} - \mathbf{s}_i\| + c \Delta t + \varepsilon_i, \quad (1)$$

where ρ_i is the measured pseudorange, $\mathbf{x}$ is the receiver position vector, $\mathbf{s}_i$ is the satellite position vector, c is the speed of light, Δt is the receiver clock bias, and ε_i represents the composite error term including atmospheric delays, measurement noise, and multipath/NLOS errors. The non-linear observation equation is linearized around an approximate position $\mathbf{x}_0$ using Taylor series expansion:

$$\Delta\rho_i = \mathbf{h}_i^\top \delta\mathbf{x} + \varepsilon_i, \quad (2)$$

where $\Delta\rho_i$ is the residual between measured and computed pseudorange, $\mathbf{h}_i = [a_{xi}, a_{yi}, a_{zi}, 1]^\top$ contains the direction cosines and clock bias coefficient, and $\delta\mathbf{x} = [\Delta x, \Delta y, \Delta z, c\Delta t]^\top$ is the state

correction vector. For n visible satellites, the system of equations in matrix form becomes:

$$\mathbf{y} = \mathbf{H}\delta\mathbf{x} + \boldsymbol{\varepsilon}, \quad (3)$$

where $\mathbf{y}$ is the observation vector of pseudorange residuals, and $\mathbf{H}$ is the design matrix. The Weighted Least Squares (WLS) solution minimizes the objective function:

$$J(\delta\mathbf{x}) = (\mathbf{y} - \mathbf{H}\delta\mathbf{x})^\top \mathbf{W}(\mathbf{y} - \mathbf{H}\delta\mathbf{x}), \quad (4)$$

yielding the normal equation:

$$\delta\mathbf{x} = (\mathbf{H}^\top \mathbf{W} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{W} \mathbf{y}. \quad (5)$$

The weight matrix $\mathbf{W}$ plays a crucial role in solution quality. In the simplest case, $\mathbf{W} = \mathbf{I}$ (identity matrix) yields the regular unweighted least squares solution. However, GNSS measurements exhibit varying quality due to satellite elevation, signal strength, and atmospheric conditions. A common elevation-dependent weighting strategy uses [4]:

$$w_i = 1/\sin^2(\theta_i), \quad (6)$$

where θ_i is the elevation angle of satellite i . This approach de-weights low-elevation satellites that are more susceptible to atmospheric delays and multipath effects. The resulting diagonal weight matrix is:

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_n \end{bmatrix}. \quad (7)$$

Alternative weighting schemes may incorporate Carrier-to-Noise density ratio (C/N_0) measurements or more sophisticated error covariance models that account for correlated atmospheric delays.

4 Code Listing

The weighted least squares is implemented in Python script as the following:

```

1 def weighted_least_squares_solution(satellite_positions,
   receiver_position, pseudoranges_meas,
2                                     satellite_clock_bias,
                                       ionospheric_delay,
                                       tropospheric_delay):
3     """Weighted least squares solution for receiver position"""
4     receiver_clock_bias = 0.0

```

```
5
6  for j in range(10): # Maximum iterations
7      # Compute geometric distances
8      estimated_distances = np.linalg.norm(satellite_positions -
9          receiver_position, axis=1)
10
11     # Correct pseudorange measurements
12     corrected_pseudoranges = pseudoranges_meas +
13         satellite_clock_bias - ionospheric_delay -
14         tropospheric_delay
15
16     # Compute residuals
17     pseudoranges_diff = corrected_pseudoranges - (
18         estimated_distances + receiver_clock_bias)
19
20     # Build design matrix G
21     G = np.zeros((len(satellite_positions), 4))
22     for i in range(len(satellite_positions)):
23         p_i = satellite_positions[i] - receiver_position
24         r_i = estimated_distances[i]
25         G[i, :3] = -p_i / r_i
26         G[i, 3] = 1.0
27
28     # Compute weight matrix
29     W = compute_weight_matrix(satellite_positions,
30         receiver_position)
31
32     # Weighted least squares solution
33     try:
34         delta_p = np.linalg.inv(G.T @ W @ G) @ G.T @ W @
35             pseudoranges_diff
36     except np.linalg.LinAlgError:
37         # Fallback to regular least squares if weights cause
38             singularity
39         delta_p, _, _, _ = np.linalg.lstsq(G, pseudoranges_diff,
40             rcond=None)
41
42     receiver_position += delta_p[:3]
43     receiver_clock_bias += delta_p[3]
44
45     # Check convergence
```

```

38         if np.linalg.norm(delta_p[:3]) < 1e-4:
39             break
40
41     return receiver_position, receiver_clock_bias

```

Listing 1: SPP weighted least-squares solver.

Where the weights are calculated through the following function (Please note here that the example code is modified to calculate weights using the formula $w_i = 1/\sin^2(\theta_i)$ instead of $w_i = \sin^2(\theta_i)$, in which the former is more commonly seen in literature):

```

1  def compute_weight_matrix(satellite_positions, receiver_position):
2      """Compute weight matrix based on elevation angles"""
3      n_sats = len(satellite_positions)
4      weights = np.zeros(n_sats)
5
6      for i, sat_pos in enumerate(satellite_positions):
7          elev_angle = compute_elevation_angle(sat_pos, receiver_position)
8          # Weight based on sine of elevation angle (higher elevation =
9              # higher weight)
10         if elev_angle > np.radians(5): # Mask low elevation satellites
11             weights[i] = 1/(np.sin(elev_angle)**2)
12         else:
13             weights[i] = 0.01 # Very low weight for low elevation
14
15     return np.diag(weights)

```

Listing 2: Weights calculation function.

5 Results and Discussion

The positioning solution from the WLS-SPP calculation is plotted in 2D in LLH coordinates and in 3D in ENU coordinates (Figure 7). From visual assessment, it is evident that the positioning solution severely deviates from the ground truth. Although this level of positioning performance is acceptable in applications such as pedestrian navigation, it would significantly compromise the safety and reliability for applications such as autonomous systems navigation.

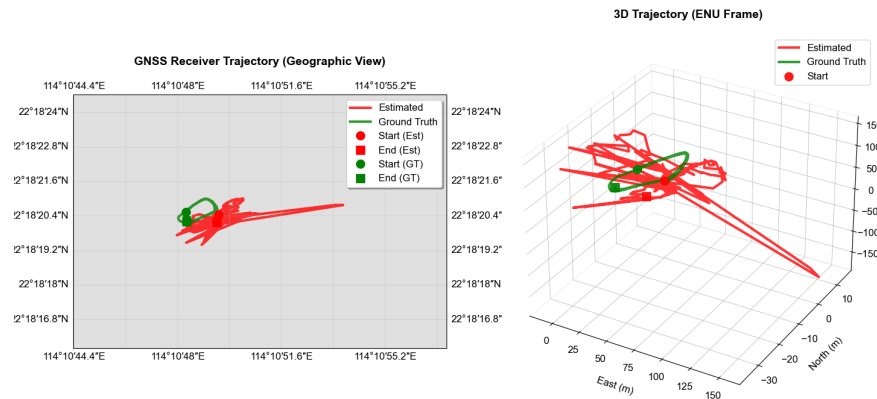


Figure 7: The trajectory solution of data collected, calculated using simple WLS SPP.

The positioning errors are then further analyzed in detail. The distributions of 2D and 3D positioning errors are illustrated in Figure 8, while the error metrics are displayed in Table 2. From the results, it can be concluded that positioning degradations from the buildings would introduce both large values of positioning error and large fluctuations in positioning accuracy (large standard deviation). The large positioning errors are most likely to be introduced from NLOS receptions, where the signals reflecting off buildings cause large delays in the ranging measurement. On the other hand, the large standard deviation may be caused by the dynamic nature of the data, in which constant changes in the environment leads to largely different positioning errors (in time instants where the environment is more open sky, the positioning error tends to be smaller, yet in time instants where the environment contains lots of obstructions, the positioning error tends to be greater).

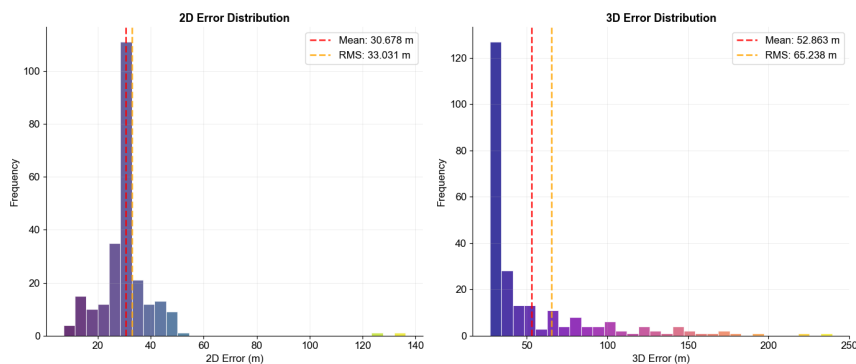
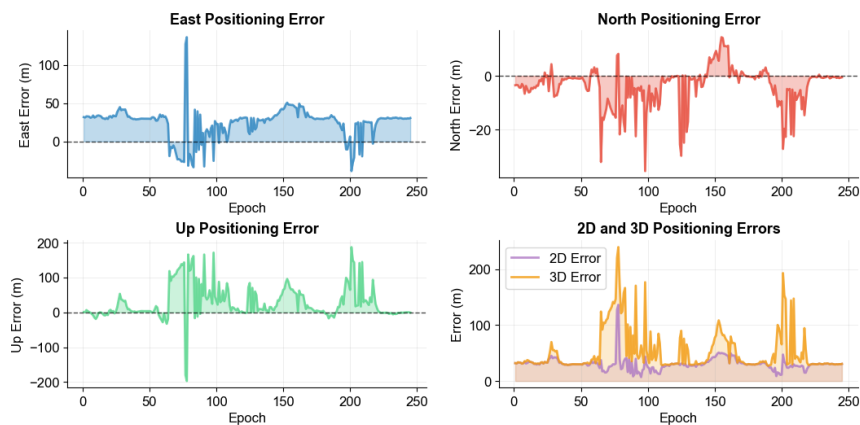


Figure 8: Distribution of 2D and 3D positioning error.

Table 2: Error metrics.

Metric	2D Error (m)	3D Error (m)
Mean	30.678	52.863
Root Mean Squared	33.031	65.238
Standard Deviation	12.243	65.328

Lastly, when plotting the positioning error along the temporal axis, it could be observed that the error tends to fluctuate and significantly increase at specific time periods. Interestingly, this time period aligns well with the time instants of lower satellite visibility identified earlier in this report. This is because lower satellite visibility limits the satellite geometry to higher DOPs, such that the final positioning error would be more sensitive to the pseudorange errors from each individual satellite.

**Figure 9:** Temporal evolution of positioning error in E, N, U, and 2D/3D directions.

5.1 So, what can we do about the positioning errors?

First of all, instead of using a simple elevation angle based weighting scheme for WLS positioning, we could combine it with other available measurements such as the carrier to noise ratio to create a more comprehensive weighting scheme [5]. Additionally, if a 3D building model of the surrounding environment is available, machine learning classifiers and ray-tracing simulations could aid in the identification of large error satellites (usually NLOS satellites) and exclude or significantly de-weight them during the position calculation process [6][7]. Sensor-fusion approaches, combining measurement inputs of GNSS, IMU, LiDAR/camera can also improve the reliability and accuracy of positioning in GNSS-challenging environments such as the urban canyon [8], while advanced hardwares such as the choke ring antenna could also aid in mitigating positioning degradations from multipath signals [9].

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